



EXAMPLE: SINGING MODEL

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0

► Suppose $h = 0$ and $J = 1$ then the potential becomes:

► We have a the Ising model has a symmetry between $U(x) = U(-x)$ and hence 2-modes

$$U(x) = - \sum_{i \sim j} x_i x_j$$

▶ We also have for lattices of dimension $d \geq 2$ the Ising model undergoes a phase transition:

► In $d=2$ the critical temperature is β_c

$$\beta_c = \frac{\log(1 + \sqrt{2})}{2} \approx 0.44$$

► $\beta < \beta_c$ we have $n(x) \approx 0$

► $\beta > \beta_c$ we have $n(x) \approx \pm n_*(\beta)$

EXAMPLE: ISING MODEL

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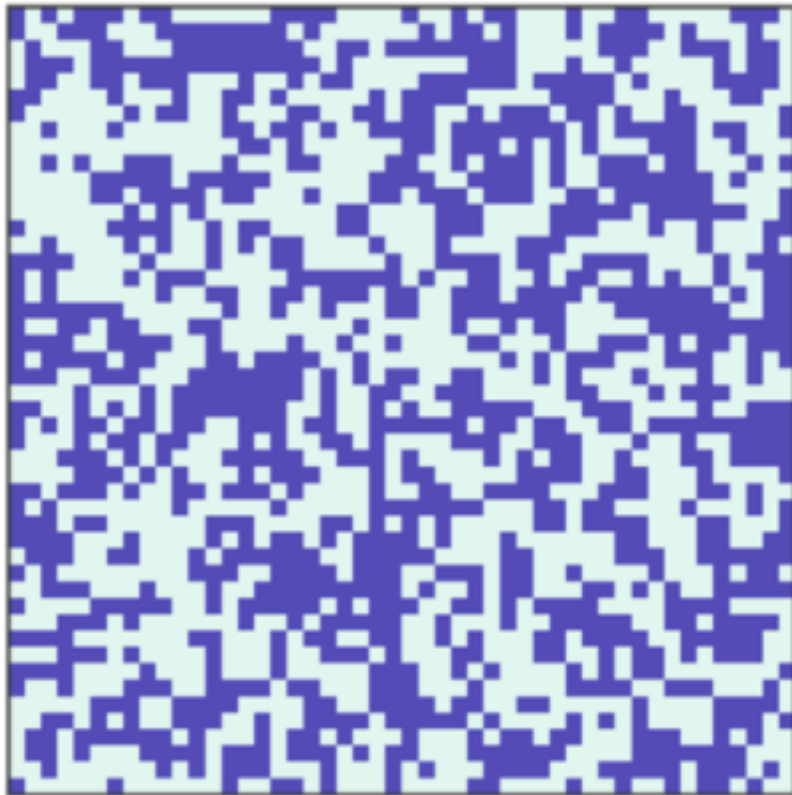
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- ▶ $\beta < \beta_c$ we have $m(x) \approx 0$
- ▶ $\beta > \beta_c$ we have $m(x) \approx \pm m_*(\beta)$

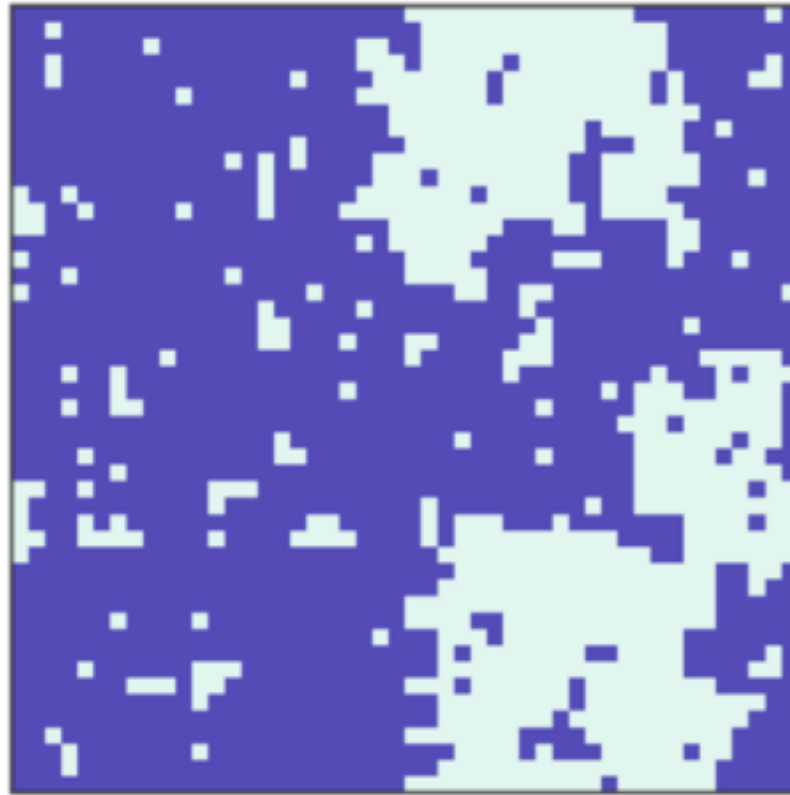
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$$\beta = 0.15$$



$$\beta = 0.44$$



$$\beta = 1.5$$

